2025 SNU FastMRI Challenge
Dataset Explanation

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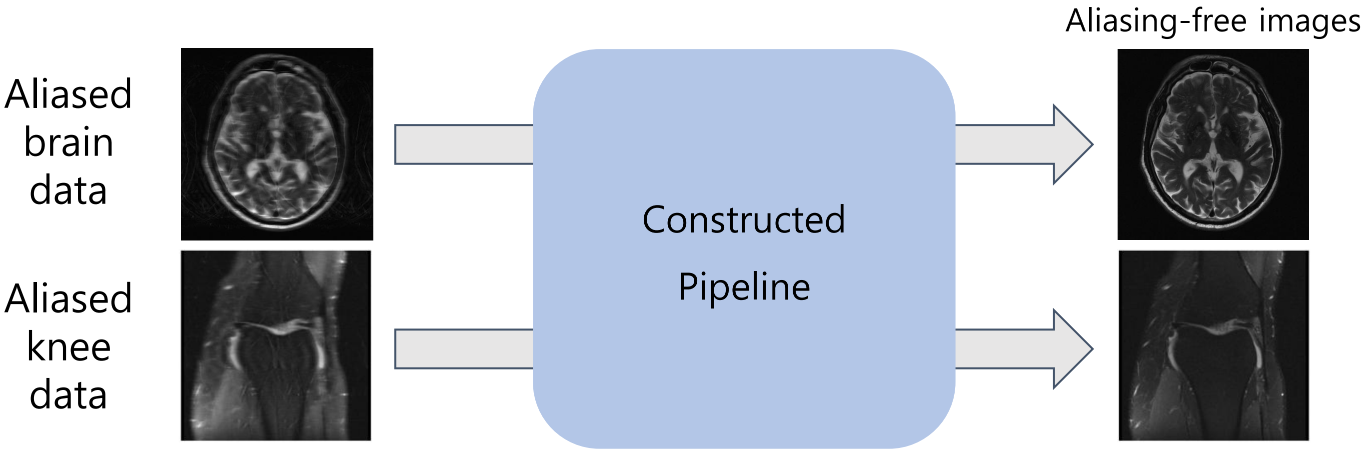
Dataset

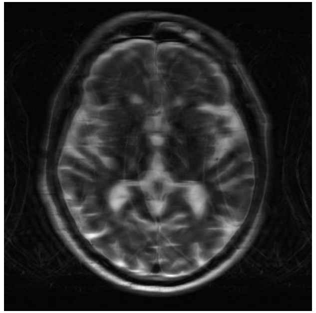
- Requires "Data Sharing Agreement" sign up
- Train/Val dataset (400 subjects, each about 20 slices)
  - brain: 100 subjects for **4x**, 100 subjects for **8x** acceleration mask
  - knee: 100 subjects for **4x**, 100 subjects for **8x** acceleration mask

- Goal: Construct **robust** pipeline across <u>different body parts</u> & accelerations

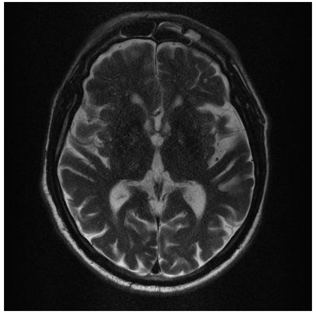


Dataset

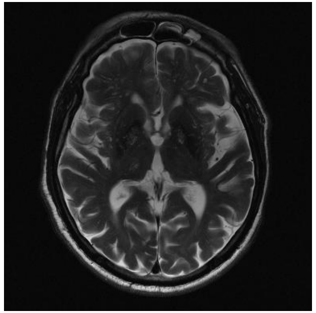
- **Train/Val** dataset (400 subjects, each about 20 slices)
- <Image>
  - Alised image, GRAPPA reconstructed image
  - Aliasing free image (ground truth)
- <Kspace>
  - Multi-coil k-space data (**full-sampled data with undersampling mask**)
- **No outside dataset allowed!!! (No pretrained model!!)**



Aliased image



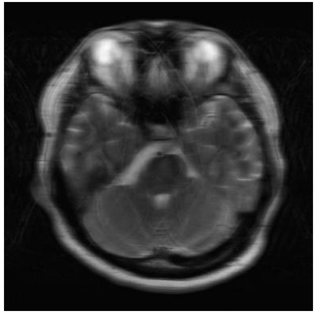
GRAPPA reconstructed image



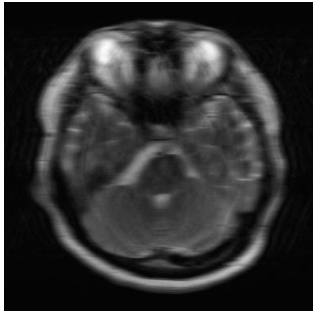
Aliasing free image (ground truth)

Dataset

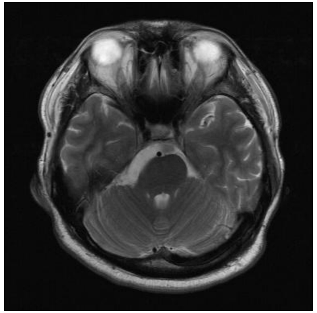
- **Test** dataset for leaderboard (116 subjects, each about 20 slices)
  - 58 subjects for brain, 58 subjects for knee
  - **4x**(29 subjects), **8x**(29 subjects) acceleration mask for each body part
  - Image: Aliased image, GRAPPA reconstructed image, Aliasing free image (ground truth)
  - kspace: Multi-coil k-space data (**undersampled data only**)
  - **Do not use for training!!!**
- **Test** dataset for final evaluation
  - <u>Will not be available for public</u>



Aliased image (**4x** acceleration mask)



Aliased image (**8x** acceleration mask)



Aliasing free image

Data format

- Data format : .h5
  - Image data

keys	\`image_input\`	RSS combined magnitude image (aliased)
	\`image_label\`	RSS combined magnitude image (aliasing free)
	\`image_grappa\`	RSS combined magnitude image (GRAPPA)
attributes	\`max\`	max value of image_label for ssim calculation
	\`norm\`	norm value of image_label

  - K-space data

keys	\`kspace\`	**full-sampled k-space data** (Train dataset) **undersampled k-space data** (Leaderboard dataset)
	\`mask\`	sampling mask used for under-sampling : 4x or 8x acceleration mask

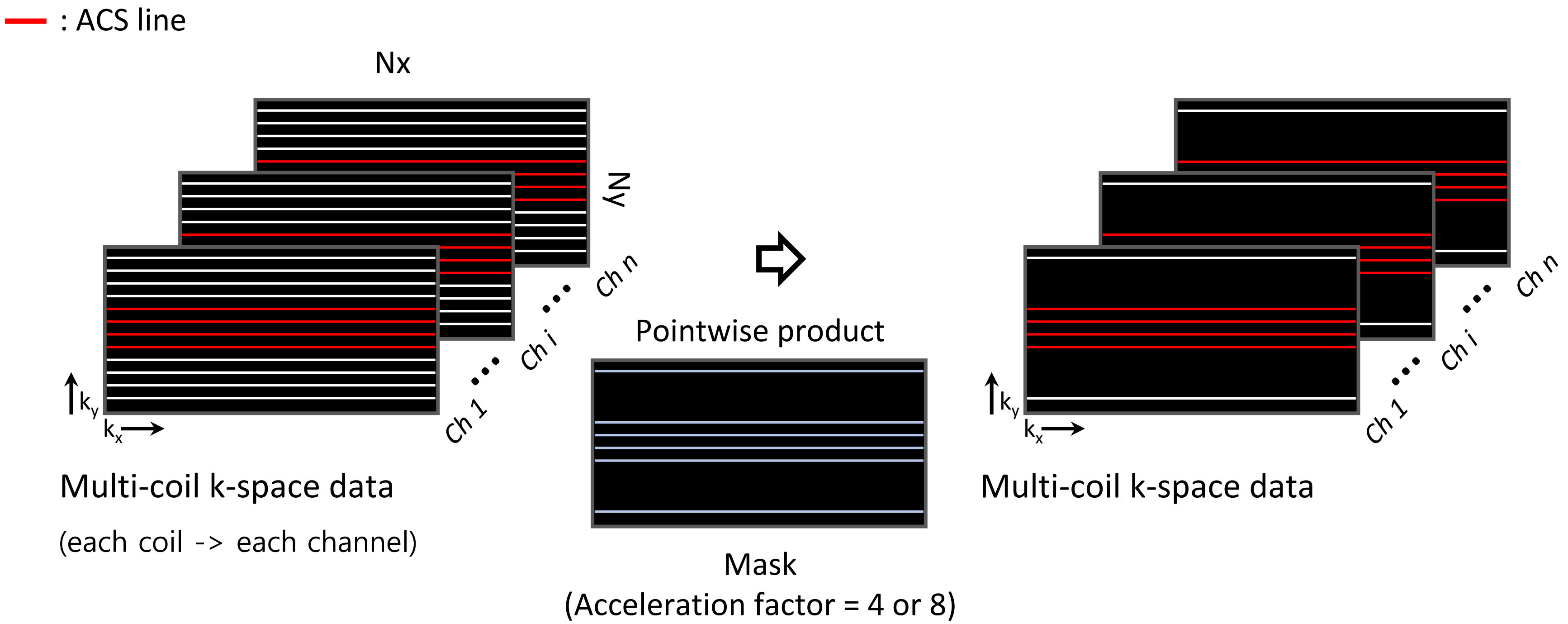
Dataset Hierarchy

- train/val
  - image
    - brain_acc4_1.h5 ~ brain_acc4_100.h5
    - brain_acc8_1.h5 ~ brain_acc8_100.h5
    - knee_acc4_1.h5 ~ knee_acc4_100.h5
    - knee_acc8_1.h5 ~ knee_acc8_100.h5
  - kspace
    - brain_acc4_1.h5 ~ brain_acc4_100.h5
    - brain_acc8_1.h5 ~ brain_acc8_100.h5
    - knee_acc4_1.h5 ~ knee_acc4_100.h5
    - knee_acc8_1.h5 ~ knee_acc8_100.h5

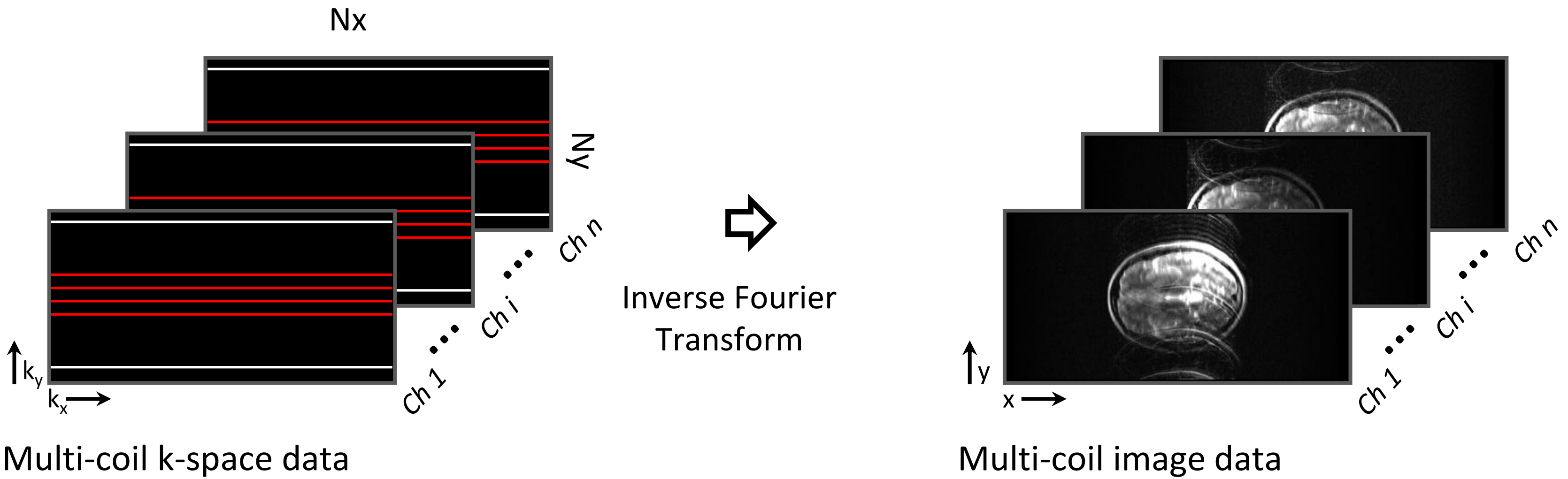
Do not utilize file/directory name to inference process!

- leaderboard
  - acc4
    - image
      - brain_test1.h5 ~ brain_test29.h5
      - knee_test1.h5 ~ knee_test29.h5
    - kspace
      - brain_test1.h5 ~ brain_test29.h5
      - knee_test1.h5 ~ knee_test29.h5
  - acc8
    - image
      - brain_test1.h5 ~ brain_test29.h5
      - knee_test1.h5 ~ knee_test29.h5
    - kspace
      - brain_test1.h5 ~ brain_test29.h5
      - knee_test1.h5 ~ knee_test29.h5

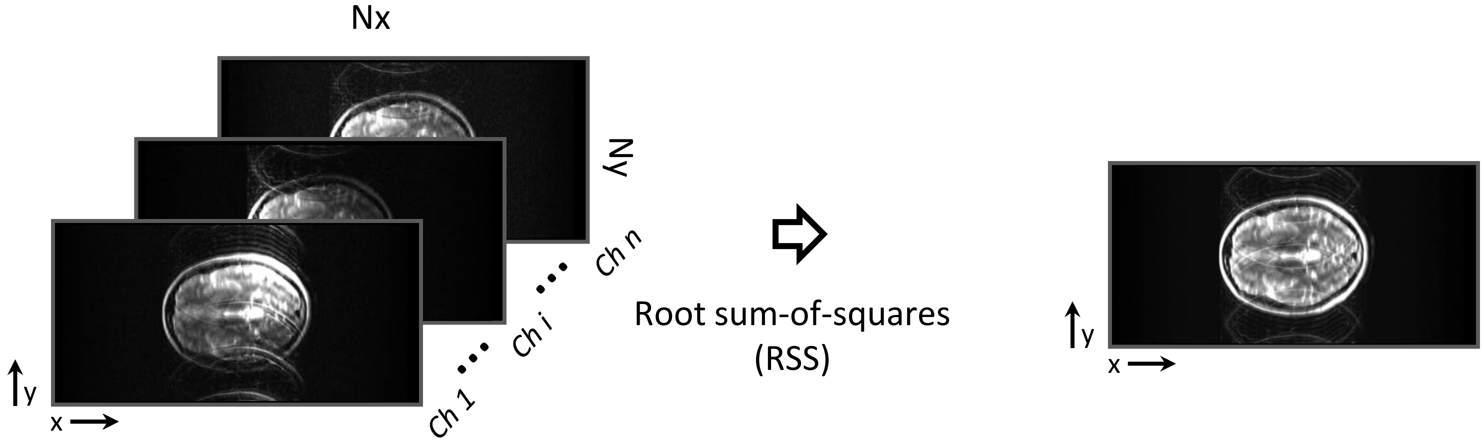
How to generate image_input from k-space data - 1 (Optional)



How to generate image_input from k-space data - 2 (Optional)



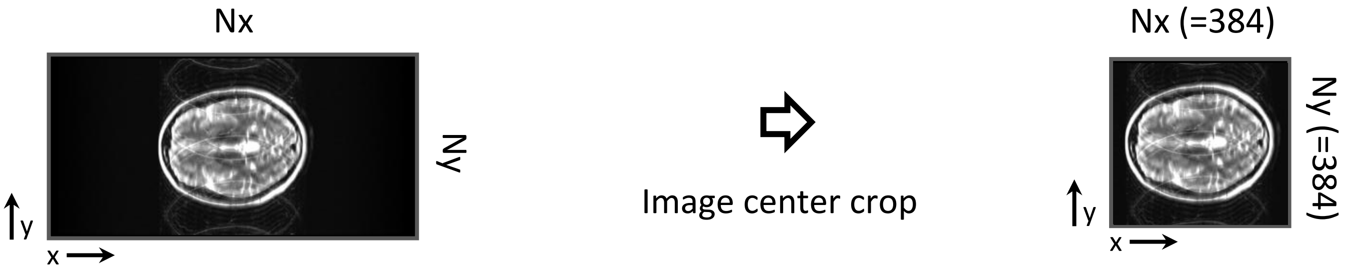
How to generate image_input from k-space data - 3 (Optional)



Multi-coil image data

Coil-combined image data

How to generate image_input from k-space data - 4 (Optional)



Coil-combined image data

Image_input

Thank you!